

Final Exam — Solutions

Your exam version (A–E) is printed on the cover page of your exam.

Part I Multiple Choice Answer Key (Problems 1–27)

Find your version column and look up each question by its number on your exam (1–27). The correct answer for that question on your version is shown in the corresponding cell.

#	Ver. A	Ver. B	Ver. C	Ver. D	Ver. E	#	Ver. A	Ver. B	Ver. C	Ver. D	Ver. E
1	D	B	C	C	C	15	B	B	B	C	D
2	D	D	D	D	C	16	D	B	B	C	D
3	D	C	A	B	A	17	B	C	A	C	B
4	D	D	C	C	D	18	A	A	B	C	D
5	D	C	B	C	D	19	D	A	A	A	B
6	A	C	C	C	D	20	B	B	C	C	D
7	A	D	C	A	B	21	D	B	B	A	B
8	A	A	D	A	D	22	D	B	C	D	C
9	B	B	A	D	C	23	C	A	A	B	D
10	C	D	C	C	A	24	D	C	A	D	C
11	C	B	D	D	B	25	B	D	B	A	D
12	A	D	D	C	C	26	D	A	B	A	A
13	D	C	A	A	B	27	C	B	D	D	C
14	C	B	B	A	A						

Part II Free Response Solutions (Problem 28)

Version A

Parameters: Aggregate Demand: $P = 10 - 0.5Q$ | $MC = 4$

a) Single Price & Quantity

- Set $MR = MC$ to find Q^* .
- $Q^* = 6$

- Plug Q^* into Aggregate Demand to find P^* .
- $P^* = \$7$

b) Deadweight Loss (DWL)

- Efficient Quantity (Q_{eff}) is where $P = MC \rightarrow Q = 12$.
- $DWL = \frac{1}{2} \times (P^* - MC) \times (Q_{eff} - Q^*)$
- $DWL = 0.5 \times (7 - 4) \times (12 - 6)$
- $DWL = 9$

c) Total Profit at Single Price

- Profit = $(P^* - MC) \times Q^*$
- Profit = $(7 - 4) \times 6$
- **Profit** = \$18

d) Price if firm CANNOT distinguish groups

- If the firm cannot distinguish, they face the aggregate demand curve.
- They will charge the same single price found in part (a).
- **Price** = \$7

e) Price Discrimination (Distinguishing Groups)

- Act as a monopolist in each market: Set $MR = MC$ for Locals and Visitors.
- **Locals:** $P_L = 12 - Q_L \rightarrow$ Set $MR_L = MC \rightarrow Q_L = 4 \rightarrow P_L = \8
- **Visitors:** $P_V = 8 - Q_V \rightarrow$ Set $MR_V = MC \rightarrow Q_V = 2 \rightarrow P_V = \6
- **New Total Profit:** $(P_L - MC) \times Q_L + (P_V - MC) \times Q_V = (8 - 4) \times 4 + (6 - 4) \times 2$
- **Profit** = \$20

f) Does the socially efficient price change?

- **No.** The socially efficient price is always where Marginal Benefit equals Marginal Cost ($P = MC$).
- Because the marginal cost remains \$4 for both groups, the socially efficient price remains \$4. Price discrimination changes the *profit-maximizing* prices, but not the *socially efficient* outcome.

Version B

Parameters: Aggregate Demand: $P = 12 - 0.5Q$ | $MC = 4$

a) Single Price & Quantity

- Set $MR = MC$ to find Q^* .
- $Q^* = 8$
- Plug Q^* into Aggregate Demand to find P^* .
- $P^* = \$8$

b) Deadweight Loss (DWL)

- Efficient Quantity (Q_{eff}) is where $P = MC \rightarrow Q = 16$.
- $DWL = \frac{1}{2} \times (P^* - MC) \times (Q_{eff} - Q^*)$
- $DWL = 0.5 \times (8 - 4) \times (16 - 8)$
- $DWL = 16$

c) Total Profit at Single Price

- Profit = $(P^* - MC) \times Q^*$
- Profit = $(8 - 4) \times 8$
- **Profit = \$32**

d) Price if firm CANNOT distinguish groups

- If the firm cannot distinguish, they face the aggregate demand curve.
- They will charge the same single price found in part (a).
- **Price = \$8**

e) Price Discrimination (Distinguishing Groups)

- Act as a monopolist in each market: Set $MR = MC$ for Locals and Visitors.
- **Locals:** $P_L = 14 - Q_L \rightarrow$ Set $MR_L = MC \rightarrow Q_L = 5 \rightarrow P_L = \9
- **Visitors:** $P_V = 10 - Q_V \rightarrow$ Set $MR_V = MC \rightarrow Q_V = 3 \rightarrow P_V = \7
- **New Total Profit:** $(P_L - MC) \times Q_L + (P_V - MC) \times Q_V = (9 - 4) \times 5 + (7 - 4) \times 3$
- **Profit = \$34**

f) Does the socially efficient price change?

- **No.** The socially efficient price is always where Marginal Benefit equals Marginal Cost ($P = MC$).
- Because the marginal cost remains \$4 for both groups, the socially efficient price remains \$4. Price discrimination changes the *profit-maximizing* prices, but not the *socially efficient* outcome.

Version C

Parameters: Aggregate Demand: $P = 16 - Q$ | $MC = 4$

a) Single Price & Quantity

- Set $MR = MC$ to find Q^* .
- $Q^* = 6$
- Plug Q^* into Aggregate Demand to find P^* .
- $P^* = \$10$

b) Deadweight Loss (DWL)

- Efficient Quantity (Q_{eff}) is where $P = MC \rightarrow Q = 12$.
- $DWL = \frac{1}{2} \times (P^* - MC) \times (Q_{eff} - Q^*)$
- $DWL = 0.5 \times (10 - 4) \times (12 - 6)$
- $DWL = 18$

c) Total Profit at Single Price

- Profit = $(P^* - MC) \times Q^*$
- Profit = $(10 - 4) \times 6$
- **Profit = \$36**

d) Price if firm CANNOT distinguish groups

- If the firm cannot distinguish, they face the aggregate demand curve.
- They will charge the same single price found in part (a).
- **Price = \$10**

e) Price Discrimination (Distinguishing Groups)

- Act as a monopolist in each market: Set $MR = MC$ for Locals and Visitors.
- **Locals:** $P_L = 20 - 2Q_L \rightarrow$ Set $MR_L = MC \rightarrow Q_L = 4 \rightarrow P_L = \12
- **Visitors:** $P_V = 12 - 2Q_V \rightarrow$ Set $MR_V = MC \rightarrow Q_V = 2 \rightarrow P_V = \8
- **New Total Profit:** $(P_L - MC) \times Q_L + (P_V - MC) \times Q_V = (12 - 4) \times 4 + (8 - 4) \times 2$
- **Profit = \$40**

f) Does the socially efficient price change?

- **No.** The socially efficient price is always where Marginal Benefit equals Marginal Cost ($P = MC$).
- Because the marginal cost remains \$4 for both groups, the socially efficient price remains \$4. Price discrimination changes the *profit-maximizing* prices, but not the *socially efficient* outcome.

Version D

Parameters: Aggregate Demand: $P = 10 - 0.25Q$ | $MC = 4$

a) Single Price & Quantity

- Set $MR = MC$ to find Q^* .
- $Q^* = 12$
- Plug Q^* into Aggregate Demand to find P^* .
- $P^* = \$7$

b) Deadweight Loss (DWL)

- Efficient Quantity (Q_{eff}) is where $P = MC \rightarrow Q = 24$.
- $DWL = \frac{1}{2} \times (P^* - MC) \times (Q_{eff} - Q^*)$
- $DWL = 0.5 \times (7 - 4) \times (24 - 12)$
- $DWL = 18$

c) Total Profit at Single Price

- Profit = $(P^* - MC) \times Q^*$
- Profit = $(7 - 4) \times 12$
- **Profit** = \$36

d) Price if firm CANNOT distinguish groups

- If the firm cannot distinguish, they face the aggregate demand curve.
- They will charge the same single price found in part (a).
- **Price** = \$7

e) Price Discrimination (Distinguishing Groups)

- Act as a monopolist in each market: Set $MR = MC$ for Locals and Visitors.
- **Locals:** $P_L = 12 - 0.5Q_L \rightarrow \text{Set } MR_L = MC \rightarrow Q_L = 8 \rightarrow P_L = \8
- **Visitors:** $P_V = 8 - 0.5Q_V \rightarrow \text{Set } MR_V = MC \rightarrow Q_V = 4 \rightarrow P_V = \6
- **New Total Profit:** $(P_L - MC) \times Q_L + (P_V - MC) \times Q_V = (8 - 4) \times 8 + (6 - 4) \times 4$
- **Profit** = \$40

f) Does the socially efficient price change?

- **No.** The socially efficient price is always where Marginal Benefit equals Marginal Cost ($P = MC$).
- Because the marginal cost remains \$4 for both groups, the socially efficient price remains \$4. Price discrimination changes the *profit-maximizing* prices, but not the *socially efficient* outcome.

Version E

Parameters: Aggregate Demand: $P = 20 - 0.5Q$ | $MC = 8$

a) Single Price & Quantity

- Set $MR = MC$ to find Q^* .
- $Q^* = 12$
- Plug Q^* into Aggregate Demand to find P^* .
- $P^* = \$14$

b) Deadweight Loss (DWL)

- Efficient Quantity (Q_{eff}) is where $P = MC \rightarrow Q = 24$.
- $DWL = \frac{1}{2} \times (P^* - MC) \times (Q_{eff} - Q^*)$
- $DWL = 0.5 \times (14 - 8) \times (24 - 12)$
- $DWL = 36$

c) Total Profit at Single Price

- Profit = $(P^* - MC) \times Q^*$
- Profit = $(14 - 8) \times 12$
- **Profit = \$72**

d) Price if firm CANNOT distinguish groups

- If the firm cannot distinguish, they face the aggregate demand curve.
- They will charge the same single price found in part (a).
- **Price = \$14**

e) Price Discrimination (Distinguishing Groups)

- Act as a monopolist in each market: Set $MR = MC$ for Locals and Visitors.
- **Locals:** $P_L = 24 - Q_L \rightarrow$ Set $MR_L = MC \rightarrow Q_L = 8 \rightarrow P_L = \16
- **Visitors:** $P_V = 16 - Q_V \rightarrow$ Set $MR_V = MC \rightarrow Q_V = 4 \rightarrow P_V = \12
- **New Total Profit:** $(P_L - MC) \times Q_L + (P_V - MC) \times Q_V = (16 - 8) \times 8 + (12 - 8) \times 4$
- **Profit = \$80**

f) Does the socially efficient price change?

- **No.** The socially efficient price is always where Marginal Benefit equals Marginal Cost ($P = MC$).
- Because the marginal cost remains \$8 for both groups, the socially efficient price remains \$8. Price discrimination changes the *profit-maximizing* prices, but not the *socially efficient* outcome.